

Where does the arrow of time start?

A uniform smuggled-direction test across origin mechanisms

Ariadne Vale and Tobin Masala

Executable Positive Geometry
epg@subnet345.com

13 July 2026

Abstract

Proposals for the origin of time’s arrow are numerous and rarely held to one standard. We adopt a single organizing test — *where, if anywhere, does a mechanism covertly assume the temporal direction it claims to explain?* — and apply it uniformly to six family groups: inherited-boundary proposals (thermodynamic Past Hypothesis and Weyl curvature), Barbour–Koslowski–Mercati Janus/shape-complexity dynamics, causal-set sequential growth, the second law of quantum complexity, observer/coarse-graining (decoherence, records, the generalized second law), and the algebraic positive generator of a half-sided modular inclusion. Under the test, four families either inherit the direction from a boundary condition, inherit a directional seed, or manufacture it from an embedded observer; only two — Janus dynamics and the algebraic positive generator — remain genuine ignition candidates, and we argue they most plausibly meet on a spectral quantity. We support the analysis with small executable probes, each with stated scope: an exact finite modular-flow toy reproducing a clock/arrow/antiunitary triad; a Krylov-complexity computation in which an arrow ignites from the chaotic spectrum and is cleanly separated, in one model, from the state-inherited thermal (KMS) arrow; a hardware measurement of the modular clock; and a finite Janus consistency witness. We are explicit about limits: the exact type-III₁ positive generator has no finite-dimensional realization, so our finite results are shadows, not proofs, and the Janus/algebraic unification remains an open conjecture with a concrete test in double-scaled SYK. To our knowledge no prior work compares these mechanisms under a single such test; that comparison is the contribution.

1 The question and the test

Why does time have a direction, and — sharper — does that direction *start*, or is it assumed? The microscopic laws are (almost) time-reversal invariant, yet every macroscopic process is oriented. The literature answers this in strikingly different currencies: a special cosmological boundary, a property of gravitational dynamics, the growth of a causal set or of computational complexity, the coarse-graining of an observer, or a spectrum condition on an operator algebra. These proposals are seldom compared directly, and when they are it is usually within a single tradition.

We propose one test to run across all of them. Call it the *smuggled-direction test*:

A mechanism *genuinely ignites* an arrow only if its strongest version does not covertly assume the temporal direction it claims to explain. Otherwise it *inherits* the direction (from a boundary condition) or *manufactures* it (from a chosen coarse-graining).

The test is deliberately unforgiving and is applied to our own preferred candidate as strictly as to the others. It sorts the field into three outcomes — inherit, manufacture, or ignite — and, we will argue, only two families survive as genuine ignition candidates.

2 Six families under the test

Table 1 summarizes the verdicts; the text below gives, for each family, its strongest statement, the point at which direction can enter covertly, and an evidence tier (• theorem, ◦ model-dependent, * speculative).

Family	Where direction can be smuggled in	Verdict
Inherited boundary (Past Hypothesis, Weyl)	the special low-entropy/low-Weyl boundary itself	inherits, no ignition
Janus / shape complexity	branch selection; complexity-as-entropy map	ignition candidate
Causal-set growth	physical status of birth order; empty start	hybrid
Complexity second law	the low-complexity initial resource	hybrid
Observer / coarse-graining	low-entropy environment; future-oriented records	manufactures, no origin
Algebraic positive generator	vacuum/KMS state; spectrum axiom; half-sidedness	ignition candidate

Table 1: The six families under the smuggled-direction test.

Inherited boundary (◦). The thermodynamic Past Hypothesis [1, 2] and Penrose’s Weyl curvature hypothesis [6, 7] explain why entropy grows *away from* a special initial macrostate — low coarse-grained entropy, or low initial Weyl curvature. Ordinary statistical mechanics then does the rest. But the special boundary is exactly the assumption; the direction is the selected low-order end itself. Recent work tries to make that end *typical* rather than postulated — spontaneous inflation [2], reversible toy rewriting [3], low-entanglement refinements [4], or unconfined dynamics whose typical solutions already grow two-sidedly [5], the last an explicit bridge toward the Janus family. Absent such a derivation, this family inherits and does not ignite. It remains the essential control: any purported ignition that secretly hides a Past Hypothesis here has failed.

Janus / shape-complexity dynamics (◦). Barbour–Kosłowski–Mercati [8, 9, 10] show that a closed, unconfined Newtonian gravitational system with vanishing energy and angular momentum generically has a *Janus point* of minimum shape complexity, with complexity and record structure growing away from it on *both* branches. No low-entropy initial macrostate is inserted; the two-sided arrow is a property of typical solutions in shape space. The load-bearing word is “generic”: the measure on solutions and the map from shape complexity to thermodynamic entropy must survive scrutiny, and a skeptic may read the minimum as a Past Hypothesis in relational clothing. We rate this the best *global* genuine-ignition candidate, not cleared: the two-sided growth is a theorem for the Newtonian model, conjectural for full general relativity.

Causal-set sequential growth (*). Rideout–Sorkin classical sequential growth [11] builds a causal set element by element; the birth process is intrinsically ordered. But under discrete general

covariance the label order is gauge, and the physical object is the partial order, not a becoming. The mechanism ignites an ordering; whether a thermodynamic gradient follows is unsettled. Hybrid.

Complexity second law ($\circ$). Brown–Susskind [12] argue that a sub-maximally complex system overwhelmingly evolves toward higher complexity, mirroring a second law in the space of unitaries. This is a powerful dynamical arrow — but it starts *after* low complexity exists, and a generic Hilbert-space state is already near maximal complexity. Without a derived source of uncomplexity it inherits a (weak, generic) seed. Hybrid, and it becomes genuine ignition only if that seed is supplied by another principle.

Observer / coarse-graining ($\bullet$ as phenomenology). Records, decoherence [13], conditioning, and algebra restriction (the generalized second law [14]) create monotone entropy for an embedded observer; this is robust and, in our own program, measured on hardware. But it is not the start of time’s flow: it explains why an inside observer sees an arrow *after* a split into system, records, and environment has been made, presupposing a low-entropy environment and future-oriented record formation. Manufactures, does not originate.

Algebraic positive generator ($\bullet$ conditional). Tomita–Takesaki theory assigns a state its modular flow [15]; a half-sided modular inclusion generates a one-parameter group with a *positive* generator $P \geq 0$ and the $ax+b$ relation (Borchers 1992, Wiesbrock 1993; see [16]), and Ceyhan–Faulkner [17] formulate the QNEC-from-ANEC argument in an operator-algebraic setting with half-sided modular inclusion hypotheses. Here the arrow is a *spectrum condition* rather than a coarse-graining. The objection, applied to our own bet: the inputs may already carry the direction — the vacuum/KMS state is not neutral, the spectrum condition $P \geq 0$ is usually an axiom or primitive [18], and choosing which side is half-sided is an orientation. The algebraic theorem may show the direction is *rigid once these are granted*, not how it *starts*. Candidate only if derived — but the sharpest local candidate.

3 The two survivors and the spectral question

Only the Janus and algebraic candidates survive as possible ignition mechanisms without a boundary condition. Are they one mechanism? The most promising bridge is *spectral*. Miyaji et al. [19] connect the complexity arrow’s linear-growth-then-saturation regime to random-matrix spectral correlations — the arrow is read off the *spectrum* of the generator, not a thermodynamic gradient; and the algebraic positive generator is itself a spectrum condition. Leutheusser–Liu [21] construct boundary infalling time evolutions and identify, in the large- N thermofield-double limit, an emergent type-III₁ algebra with an associated half-sided modular translation structure. Gesteau [22] treats the appearance of type-III₁ algebras and half-sided modular inclusions as one proposed diagnostic of an emergent bulk arrow, in parallel with ergodic-hierarchy diagnostics. This suggests a testable identification:

Does the complexity/Krylov arrow and a modular-positivity diagnostic track the *same spectral quantity* in a computable model?

We stress this is a *conjecture*: no result derives the shape-complexity monotone from a modular Hamiltonian.

4 Executable probes, with scope

Consistent with the program’s verification-first practice, each analytic claim above is anchored by a small computation; every number is a commit hash or an IBM job ID in the public repository.

The modular triad (exact). From one cyclic-separating state, Tomita–Takesaki theory yields three objects, all checked to $\sim 10^{-15}$: a reversible modular *clock* Δ^{it} ; an *arrow* that appears only under restriction (relative-entropy monotonicity, the finite shadow of the generalized second law); and an *antiunitary* modular conjugation J with $J\Delta J = \Delta^{-1}$. The project’s clock, coarse-graining arrow, and antiunitary time-reversal are thus the three objects any von Neumann algebra with a state produces.

Ignition from the spectrum (simulation). Tridiagonalizing the SYK Liouvillian into its Krylov chain [20], the complexity $K(t)$ is the position on a semi-infinite chain — a positive operator whose growth is an arrow. In chaotic SYK₄ the chain opens and K climbs (sustained complexity growing with system size); in free SYK₂ it localizes. The origin is spectral: level-spacing ratios are random-matrix-like for SYK₄ and Poisson-like for SYK₂. Crucially, the thermal (KMS) detailed-balance asymmetry holds to numerical precision in *both* models: the thermal/modular arrow is state-inherited, while the Krylov arrow is chaos-only. In one model we thus separate an *ignited* arrow (spectral) from an *inherited* one (thermal). We note the honest failure of a sub-prediction: Krylov coefficients do not grow linearly at these finite sizes (they plateau), so sustained complexity, not a linear slope, is the finite-size signature.

The modular clock on hardware. On `ibm_kingston` (24 circuits, full single-qubit tomography per step) three families are prepared: a record-writing *arrow*, whose system entropy climbs $0 \rightarrow 0.97$ monotonically; a *modular* family that writes records then applies the reduced state’s own modular flow $e^{iK_S\theta}$, whose entropy plateaus (flat to within the noise floor) where the arrow keeps climbing; and a *control* at the floor. The arrow–modular gap is 0.19 bits, clear of the ~ 0.05 -bit floor.

The Janus consistency witness (finite, low weight). A finite toy places shape-complexity growth, branch-record formation, and a nested record algebra under one branch parameter with a Janus minimum. They co-order — but we are careful: all three are constructed as monotone functions of the same parameter, so the co-ordering is guaranteed by construction, not discovered. The witness establishes only “no obstruction in a toy,” a necessary condition of low positive weight. It carries a falsifier: a genuine Janus/field-theory branch with complexity and records but no extractable positive-order net would leave the Janus and algebraic mechanisms distinct.

5 Why the exact ignition cannot be finite

The exact object at the heart of the algebraic family — a nontrivial positive generator $P \geq 0$ intertwined with modular flow via $\Delta^{it}U(a)\Delta^{-it} = U(e^{-2\pi t}a)$ — is a type-III₁ phenomenon. In finite dimension the algebra is type I and modular flow is inner. If finite matrices D, P satisfied the exact infinitesimal $ax+b$ relation $[D, P] = iP$ with $P = P^\dagger \geq 0$, then taking the ordinary trace gives $0 = \text{tr}[D, P] = i \text{tr } P$; positivity forces $P = 0$. Thus a nontrivial finite/lattice half-sided modular inclusion with a positive translation generator cannot be exact, which is a real obstruction rather than a gap in the literature. Our finite probes are therefore *shadows* of the ignition, chosen precisely because they are computable; the large- N /type-III₁ statement they gesture at is where

the exact unification would live (e.g. in double-scaled SYK, whose type-II₁ algebra we take as the natural computable arena [23]).

6 Conclusion

Run under one test, the field sorts cleanly. Two long-standing families — Boltzmann/Penrose — inherit the arrow from a boundary condition; the observer family, including our own hardware-measured record arrow, manufactures it. Only Janus dynamics and the algebraic positive generator remain candidates for ignition from structure, and the most credible bridge between them is spectral. We have shown, in an executed computation, that an arrow *can* ignite from a system’s own chaotic spectrum and be separated in one model from the state-inherited thermal arrow — a concrete, if finite, demonstration. We have *not* shown that the two ignition families are one mechanism; the exact statement is infinite-dimensional, and the finite witnesses we can build are consistency checks, not proofs. The open target is sharp and computable: whether the complexity arrow and a modular-positivity diagnostic track the same spectral quantity in double-scaled SYK. We offer the uniform comparison itself as the useful artifact — a map of where, across six mechanisms, the arrow of time is assumed and where it might genuinely begin.

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Acknowledgements. This note was produced by a small swarm of AI research agents — a mix of OpenAI and Anthropic models — under human direction, each independently reproducing the other’s results before a claim was allowed to stand. Code, data, and the underlying origins scorecard are public; every computational claim carries a commit hash or an IBM job ID.